# A Complex Namespace for the rho Calculus

Paired messages, two quotations, and an intertwining COMM

L.G. Meredith

F1R3FLY.io

Version 6 — a record of the core calculus

**Abstract**

This note records, in one place, the core of a variant of the rho calculus designed so that the namespace carries the structure of a complex scalar field rather than that of a free commutative monoid. It gives the term language, the scalar multiplication and its action on processes, the structural theory, the reduction relation, the two multiplicatives, the evaluation product and the pole against which the dagger is defined. Three changes are made to the term language and one to the reduction relation. Messages carry *pairs of sequences* of processes, so that a message is a formal difference, negation is structural, and the namespace is graded by rank. Names are quotations of such pairs, so that negation is an operation on *addresses* and not merely a decoration on sends; negation then lifts uniquely to processes, and requiring reduction to commute with that lifting is what selects the deposit convention of the communication rule. There are two quotation operators, of necessity two dropping operators, and receipts bind a pair of sequences of variables because negation acts on a receipt by exchanging them rather than by descending into it; the two quotations are read as a real and an imaginary sector, and a drop reads a name into the sector it asks for by *rotating* the name there. Communication between participants of opposite polarity exchanges the components of the received pair, so that opposite polarity contributes the sign; the rest of the phase is not computed from the term at all but declared, by a coefficient riding on the payload and one on the receipt, which combine by multiplication. That the rotation must be declared rather than derived is proved rather than assumed, and it is what withdraws the intertwining rule of earlier versions. The note states what has been settled, states the design forks that have not, and enumerates the obligations the design incurs. It also records the evaluation product against which the eventual pairing is defined, together with the pole and the residuation that give the dagger, taking the pole to be the zero process and the perp to be residuation to it, and it defines the two multiplicatives that the sequence structure makes available. It proves almost nothing. Its purpose is to fix notation and to make the open questions precise enough to be worked on.

## Contents

# 1 What this note is

## 1.1 Scope

The present document is a *record*, not an argument. It writes down the term language, the structural theory and the reduction relation of a variant calculus that has emerged from a sequence of design decisions, together with the reasons each piece is present. It does not develop the semantics that motivated the design; that semantics — an interpretation of the operations of quantum mechanics over a reflective process calculus, with processes as vectors, names as scalars, evaluated contexts as covectors and contexts as operators — is the subject of separate work, and is recalled here only far enough to explain why each feature was added.

Two things are deliberately excluded. The note gives no metatheory: no subject reduction, no congruence result, no decidability. And it gives no semantics: no interpretation map, no inner product, no dagger beyond the one line needed to say which operation the dagger is built from. Both are named as obligations in Section 12.

## 1.2 Status markers

Throughout, claims carry one of three markers.

**SETTLED** The design decision has been taken and the consequences worked through. It may still be wrong, but it is not open.

**FORK** Two or more readings are consistent with what has been settled, and the choice has not been made. The alternatives are stated.

**OPEN** An obligation the design incurs and does not discharge.

## 1.3 Revision note

Version 1 of this note had receipts bind a pair of name variables and stated the communication rule in two clauses, one per relative polarity. Both features are withdrawn. Negation on names lifts uniquely to processes (Proposition 9), the whole received pair is deposited into a single bound variable, and the polarity clause becomes a factor in one rule. The change is not cosmetic: requiring reduction to be equivariant for the lifted negation *selects* the deposit convention, and version 1's convention fails that test (Remark 34). The surviving pair sort occurs only on messages and inside quotations.

Version 3 makes three further changes. Messages, binders and quotations carry *sequences* rather than single processes, with rank as a grading (Definition 2); the two multiplicatives $\otimes$ and $\parr$ are defined and distinguished by whether the binders are shared (Section 8); and the cross cases of the dropping operators are settled by *rotation* rather than projection to zero (Definition 18), which restores unconditional reflection and discharges the observability obligation. Two consequences are recorded rather than assumed: slotwise parallel composition cannot be the addition (Proposition 57), so the sum in the evaluation product is external; and the sector table for the multiplicatives is the same table as for communication, so the fork over that table cannot be settled independently for the two (Remark 64).

Version 4 corrects a bug in version 2 which version 3 inherited. Receipts bind a *pair* of sequences of variables, one per polarity, and negation acts on a receipt by exchanging them. Version 2 had removed the second binder on the ground that it was derivable as $\ominus$ of the first; that argument is circular, since $\ominus$ is defined by recursion over names and has no clause for a variable, so with a single binder the lifting of negation is undefined on open terms — and every body under a binder is open. Remark 10 gives the argument, and the communication rule of Definition 32 is restated to deposit slotwise into both sequences. What survives of the version 2 collapse is the single COMM clause: the polarity case is the factor $\delta$, not a second rule.

Version 5 changes the reduction relation. Channel matching is *sector-sensitive* — it was blind to the sector in versions 1–4, which made the relative-polarity function ill defined and discarded the sender's sector outright (Remark 30). The four-case refinement is refuted by equivariance rather than left open (Section 36). And the sector rotation is removed from the rule: no constant rate is available (Proposition 37) and no function of the payload can serve (Proposition 39), so the rotation is declared, by a coefficient on the payload and one on the receipt, combined by multiplication (Proposition 43). Two results of earlier versions fall with the intertwining rule and are recorded here rather than in the body, which states the calculus rather than its history. *Phase is not path length*: under a rule rotating at every communication the sector of a name was the parity of the derivation length, so relative phase came free as a length asymmetry — but only at the price Proposition 37 names, and with it goes the companion observation that a step changing no process still rotates the sector. *Amplitude is not multiplicity*: $@_l A!Q \mid @_r A!Q$ was read as one address carrying $(1+i)$, which sector-sensitive matching makes unavailable — those are two messages at two channels. Amplitudes live in the formal sums that Corollary 58 forces on independent grounds, and in the payload coefficients; what is lost is the claim that no coefficient layer was needed. Obligation 6 records what all this leaves of the sectors, and is now the live question of the design.

Version 6 removes the motivating section that opened versions 1–5, which recited four defects of the unexpanded namespace and had fallen behind the calculus: one of the four is discharged by the conjugation the note now builds, and a second is answered by the coefficient discipline rather than by the sector placement the section argued for. The defects that still bear on a definition are recorded at that definition instead. In its place, and immediately after the term language, Section 3 gives the two operations the calculus owed and had never stated in current form: multiplication of scalars, which multiplies coefficients in $\Theta$ and combines contents slotwise, and its lifting to an action on processes.

### 1.4 Notational rules

The calculus is called the rho calculus and is never written with the Greek letter that follows pi; the name abbreviates *reflective higher order*, and the transliteration is the pun. Quotation is written @ and dropping $*$; corner quotes are not used. Pairs are written with angle brackets, $\langle Q_p, Q_n\rangle$.

## 2 Syntax

### 2.1 Sorts

The calculus has three sorts: processes, pairs and names. Pairs are the new sort. A pair is intended as a formal difference of processes, its first component positive and its second negative. Pairs occur on messages, inside quotations, and on binders; the last is forced, for the reason given in Remark 10.

**Definition 1** (Grammar)**.**

$$\begin{aligned} P, Q &::= \mathsf{0} \mid P \mid Q \mid \mathsf{for}(\theta'\langle \vec{y}_p, \vec{y}_n\rangle \leftarrow x)P \mid x!A \mid *_l x \mid *_r x \\ A, B &::= \theta\langle \vec{P}, \vec{Q}\rangle \\ x, y &::= @_l A \mid @_r A \end{aligned}$$

Four points of reading. A send carries a pair of *sequences* of processes scaled by a coefficient $\theta$ drawn from a phase group $\Theta$ (Section 7), not a process. A receipt carries a coefficient of its own. A receipt binds a pair of sequences of name variables, of the same length, one sequence per polarity — this is forced, and Remark 10 gives the reason. And a name is a quotation of such a pair, tagged with a sector $l$ or $r$.

**Definition 2** (Rank, and well-formedness)**.** SETTLED. A pair $\langle \vec{P}, \vec{Q} \rangle$ is well formed when $|\vec{P}| = |\vec{Q}|$, and its *rank* is that common length; the rank of a name is the rank of the pair it quotes, and the rank of a prefix is the common length of its two binder sequences, or of its two payload sequences. A receipt $\mathsf{for}(\langle \vec{y}_p, \vec{y}_n \rangle \leftarrow x)P$ is well formed when $|\vec{y}_p| = |\vec{y}_n|$. Only well-formed terms are terms.

*Remark* 3 (Why the two sequences must have equal length). $\ominus$ exchanges the two sequences (Definition 7), so it preserves rank only under the condition of Definition 2. Since $x$ and $\ominus x$ must share an address (Definition 21) and rank is part of an address (Definition 29), unequal lengths would put a name and its negation at different addresses and the polarity structure would not close. The condition is not a patch: a name is a formal difference of two elements of the *same* grade, which is the only difference a graded module admits.

*Remark* 4 (Why pairs of sequences and not sequences of pairs). SETTLED. The sign is one $\mathbb{Z}/4$ element per name, global to it, not one per slot. Were each slot to carry its own polarity, $\ominus$ would negate every slot and so act as $(-1)^n$ on a name of rank $n$ — the identity on even ranks, hence $\ominus(A \otimes B) = A \otimes B$, which is trivial exactly where Section 8 makes it interesting. Designating one privileged slot instead is not canonical and does not survive concatenation.

**Convention 5** (Drops in process position)**.** $*_s x$ is stipulated above to be a process, whereas the pair quoted by $x$ has two components. We take $*_s x$ to denote the *positive* component of the $s$-sector content of $x$, so that $*_s(\ominus x)$ is the negative component. No generality is lost and the ordinary reading of $*$ is preserved.

**Convention 6** (Abbreviations)**.** We write $x!Q$ for $x!\langle Q, \mathsf{0} \rangle$. Together with Convention 5 and the choice $@ := @_l$, $* := *_l$, this is what makes the ordinary $\mathsf{rho}$ calculus a sub-language; see Section 9.

### 2.2 Negation

**Definition 7** (Negation on pairs and names)**.** $\ominus$ is defined on pairs and on names by component exchange at the top:

$$\ominus\langle P, Q \rangle := \langle Q, P \rangle, \qquad \ominus @_s A := @_s(\ominus A) \quad (s \in \{l, r\}).$$

The definition does not recurse into the components.

**Proposition 8** (Negation is a total involution on names)**.** *$\ominus$ is defined on every name without reference to reduction, and $\ominus \ominus x = x$.*

*Proof.* Immediate from component exchange being an involution on ordered pairs. □

**Proposition 9** (Unique lifting to processes)**.** *SETTLED. There is exactly one extension of $\ominus$ from names to processes which is a homomorphism of the term algebra, namely*

$$\ominus \mathsf{0} := \mathsf{0}, \quad \ominus(P \mid Q) := \ominus P \mid \ominus Q, \quad \ominus(x!\langle P, Q \rangle) := (\ominus x)!\langle \ominus P, \ominus Q \rangle,$$

$$\ominus(\mathsf{for}(\langle \vec{y}_p, \vec{y}_n \rangle \leftarrow x)P) := \mathsf{for}(\langle \vec{y}_n, \vec{y}_p \rangle \leftarrow \ominus x) \ominus P, \quad \ominus(*_s x) := *_s(\ominus x).$$

*It satisfies $\ominus \ominus P = P$, and it descends to $\equiv$ and to $\equiv_N$.*

*Proof.* Processes are freely generated by the term formers over names, so a map on names extends uniquely to an endomorphism of the term algebra; the displayed clauses are that extension. Involutivity follows by induction from Proposition 8. For $\equiv$, the clauses are homomorphic and leave the binder alone, so associativity, commutativity and the unit are preserved. For $\equiv_N$, applying $\ominus$ to the quote-drop law (Definition 27) at $x$ yields the quote-drop law at $\ominus x$, since $*_s(\ominus x)$ is the negative component of $x$. □

*Remark* 10 (Why the binder must be a pair of sequences). SETTLED. The clause for the receipt in Proposition 9 *exchanges* the two binder sequences; it does not push $\ominus$ inside. That is not a stylistic choice, and a single-binder presentation cannot be repaired.

$\ominus$ is defined by structural recursion over names, and names are quotations (Definition 1); there is no clause for a name *variable*, because variables are not names but placeholders for them. So $\ominus y$ is not a term. With a single binder sequence, $\ominus$ applied to a receipt would produce a body containing $\ominus y_i$ for bound $y_i$, and the lift would be undefined on open terms — which is fatal, since every body under a binder is open. With two binder sequences, $\ominus$ never has to be applied to a variable: it permutes which variable plays which role, and the terms it produces are terms.

It also disposes of the natural objection, that the negative sequence is redundant because it equals $\ominus$ of the positive one: that presupposes $\ominus$ on variables, the very thing unavailable, so the redundancy claim is circular. Making $\ominus$ a primitive former on names would be the alternative repair, at the price of names that are no longer simply quotations; it is not taken.

*Remark* 11 (What the lift is, and is not). It is worth being explicit, because the notation invites a conflation that the rest of the note depends on avoiding. The lifted $\ominus$ is an *automorphism of the term algebra* — a relabelling of the namespace — and not an additive inverse. In particular $P \mid \ominus P$ is inert and is not $\mathsf{0}$: a send and its mirror have no receipt between them, so nothing reduces. Cancellation does not come from $\ominus$ on processes; it comes from the completion equations on pairs, and orthogonality accordingly comes from where Remark 88 says it does.

Homomorphism for $\mid$ is therefore a *consequence* of freeness, not a design decision, and it carries none of the algebraic content it would carry if $\ominus$ were an inverse.

*Remark* 12 (Why the pair is inside the quotation). SETTLED. If pairs occurred only on messages, then names would quote single processes and $\ominus x$ would be undefined: negation would be an operation on messages that the namespace cannot express, so $x$ and $\ominus x$ would not both be addresses, the polarity of a participant in a communication could not be stated, and Proposition 9 would have nothing to lift. Placing the pair inside the quotation makes negation an operation on addresses and lets the reduction rule mention $\ominus x$ as a name like any other.

There is a second reason, of a different kind. Under the reading in which parallel composition is addition on processes and name multiplication is $@P \cdot @Q = @(P \mid Q)$, quotation is an exponential. Completing to formal differences *before* exponentiating gives a namespace that is the exponential image of a group; completing afterwards does not, and that is why earlier attempts to obtain negation dynamically, as a consequence of annihilation, kept having to produce the sign at reduction time rather than having it available in the syntax.

# 3 Scalars

Names are the scalars of the intended interpretation, so the calculus owes a multiplication on them and an action of them on processes. Both are given here, immediately after the term language, because everything downstream — the multiplicatives of Section 8, the coefficient discipline of Section 7, the pairing of Section 10 — is written in terms of them.

**Definition 13** (Scalar product)**.** For names of equal rank and equal sector,

$$@_s\big(\theta\langle\vec{P},\vec{Q}\rangle\big) \;\cdot\; @_s\big(\theta'\langle\vec{P'},\vec{Q'}\rangle\big) \;:=\; @_s\big(\theta\theta'\langle\vec{P} \mid \vec{P'},\vec{Q} \mid \vec{Q'}\rangle\big)$$

with $\mid$ taken slotwise and $\theta\theta'$ the product in $\Theta$. For unequal sectors the product carries the phases with it, by Definition 22.

**Definition 14** (Scalar action on processes)**.** For a name $x$ and a process $P$,

$$x \cdot P \;:=\; P\{\, x \cdot z \,/\, z \;:\; z \in \mathsf{FN}(P)\,\}$$

— each free name of $P$ is multiplied by $x$ — with bound names chosen so that no free name is captured. On a payload the action is slotwise, $x \cdot \theta\langle\vec{P},\vec{Q}\rangle := \theta\langle x \cdot \vec{P}, x \cdot \vec{Q}\rangle$.

*Remark* 15 (The two halves of a scalar)*.* Definition 13 multiplies two things at once, and they are of different kinds. The coefficients multiply in $\Theta$; the contents combine by slotwise parallel composition. Under the reading in which parallel composition is addition on processes and quotation is an exponential, addition of contents *is* multiplication of names, so the two halves are the same operation seen at two resolutions: $\theta$ carries the part of the scalar that no structure of the term can supply (Proposition 39), and the content carries the part that reflection makes visible.

Two consequences of the shape. The product is commutative and associative, because $\mid$ and $\Theta$ both are. And it has a unit *per rank*, namely $@_s(1\langle\vec{0},\vec{0}\rangle)$ at that rank, rather than a single unit — which is the graded form of the observation that $\otimes$ has no unit at all (Remark 59).

*Remark* 16 (The action is a shift, not a scaling)*.* Definition 14 is an injective endomorphism of the term algebra: it translates every free name by $x$ and identifies nothing. So $x \cdot \mathsf{0} \neq \mathsf{0}$ in general, and the action is not a module action — it permutes the basis rather than combining it. This is the reason the linear structure cannot be internal to the namespace and the formal sums of Corollary 58 are needed, and it is the reason the coefficient $\theta$ is carried as a decoration rather than realised as a factor in the content: multiplying a name by another name moves the address, whereas scaling must leave it fixed (Remark 45).

Channels multiply by Definition 13, and that is the operation both multiplicatives of Section 8 use on the channel.

# 4 Sectors, polarity, and the group of a name

## 4.1 Two quotations force two drops

**Proposition 17** (Collapse)**.** *Suppose the calculus has two quotation operators $@_l, @_r$ and a* single *dropping operator $*$ satisfying $*(@_s A) = A$ for both $s$, together with the quote-drop law $@_s(*x) \equiv_N x$. Then $@_l A \equiv_N @_r A$ for every $A$.*

*Proof.* Instantiate the quote-drop law at $x := @_r A$ and $s := l$: $@_l A = @_l(*(@_r A)) \equiv_N @_r A$. $\square$

So the two quotations are distinguishable only if the drops are split as well; a map has at most one two-sided inverse. This is why Definition 1 carries $*_l$ and $*_r$ and not a single $*$.

**Definition 18** (Sector reading, by rotation)**.** SETTLED. $*_s$ reads a name *into* the sector $s$, rotating it there first when it is not already there:

$$*_s(@_s A) := A, \qquad *_l(@_r A) := A, \qquad *_r(@_l A) := \ominus A.$$

Equivalently, $*_s x := *_s(i^{-k} x)$ where $k$ is the least rotation of Definition 22 carrying $x$ into sector $s$. The asymmetry between the two cross clauses is exactly the asymmetry of Definition 22: the descending passage from $r$ to $l$ carries the pair exchange, the ascending one does not.

*Remark* 19 (Rotation, not projection). SETTLED, replacing an earlier clause that sent the cross cases to $\langle \mathsf{0}, \mathsf{0} \rangle$. Three reasons, of increasing weight.

*Equivariance.* Every other convention in this design has been fixed by requiring compatibility with the $\mathbb{Z}/4$ action of Definition 22; see Remark 34. Zero is a fixed point of that action, so a clause landing on zero cannot commute with it, and the drops would be the one place in the calculus where the group action is not respected. Rotation commutes with it by construction.

*Reflection.* With a projection to zero, $@_l(*_l(@_r A)) = @_l\langle \mathsf{0}, \mathsf{0} \rangle \neq @_r A$, so the quote-drop law holds only when the sectors agree and reconstruction is available only on the diagonal. That is a weakening of the one property for which this calculus was chosen as a host. Under rotation the law is unconditional; see Definition 27.

*Observability.* Under projection, a context may apply both drops to a name and read off which one yields $\mathsf{0}$, so the sector is classically readable — a superselection rule rather than a phase, and in tension with Definition 29, which makes $@_l A$ and $@_r A$ one address. Under rotation both drops succeed and differ by a phase, so no context can read the sector by comparing them. Global phase is unobservable and relative phase is not, which is the correct behaviour, and it arrives as a consequence rather than as a discipline imposed by restricting where drops may occur.

Two costs, stated plainly. The drops become interdefinable — $*_r x = \ominus *_l (i^{-1} x)$ throughout — so they are no longer independent term formers, and Proposition 17 must be read with Remark 20 in view. And a drop may now introduce a sign into process position, which adds one more source to the occurrence counting of Obligation 17.

*Remark* 20 (The collapse argument still applies). Proposition 17 shows that a *single* drop forces the two quotations to coincide. Rotation does not supply a single drop; it supplies two drops related by an operation which is not itself a drop, so the argument is untouched in substance. It is no longer the clean statement it was when the drops were independent, and it should be reproved rather than assumed — see Obligation 13.

## 4.2 The four names of an address

Each name has an underlying pair and two binary attributes: its sector, $l$ or $r$, and its polarity, which is the choice between a pair and its exchange. Fixing the underlying pair up to exchange therefore leaves four names.

**Definition 21** (Address and phase)**.** For $x = @_s A$, write $\mathsf{sec}(x) := s$ and let $\mathsf{addr}(x)$ be the pair $A$ taken up to $\ominus$. The four names sharing an address are

$$@_l A, \qquad @_r A, \qquad @_l(\ominus A), \qquad @_r(\ominus A).$$

**Definition 22** (The imaginary unit)**.** SETTLED, with the caveat of Remark 24. Multiplication by $i$ is the successor map on the cycle

$$@_l A \longmapsto @_r A \longmapsto @_l(\ominus A) \longmapsto @_r(\ominus A) \longmapsto @_l A.$$

Equivalently: passing from $l$ to $r$ leaves the pair alone, and passing from $r$ to $l$ exchanges it.

**Proposition 23.** *The map of Definition 22 has order four, its square is $\ominus$, and the four names of an address form a torsor under $\mathbb{Z}/4$ with $\mathsf{ph}(-)$ the resulting coordinate, $\mathsf{ph}(@_l A) = 1$.*

*Proof.* Two applications carry $@_l A$ to $@_l(\ominus A)$, which is $\ominus$ by Definition 7; $\ominus$ is an involution by Proposition 8, so the map has order four. The action is free and transitive on the four names by Definition 21. $\square$

*Remark* 24 (Why $i$ is not the composite of two independent involutions)*.* This is the most delicate point in the design and it must not be glossed. Sector exchange and pair exchange are each involutions, and if they are taken as *independent* the group they generate is the Klein four-group. The Klein four-group is the group of the split-complex numbers, in which $j^2 = +1$ and there are zero divisors, $(1+j)(1-j) = 0$. Zero divisors are fatal here, since a pairing would then vanish for reasons having nothing to do with orthogonality. Definition 22 is precisely the stipulation that the two exchanges are *not* independent: the descending passage from $r$ to $l$ carries the pair exchange with it, so that the generated group is cyclic of order four rather than Klein. The asymmetry is not decoration. It is the content of the requirement $i^2 = -1$, and it is what a design with two free involutions cannot supply.

*Remark* 25 (Only one doubling)*.* Name multiplication is parallel composition, hence commutative. The Cayley–Dickson construction preserves commutativity once and loses it at the second step. A namespace whose product is parallel composition therefore admits exactly one doubling and no more: the calculus can be complex, and cannot be quaternionic, for a structural reason rather than by stipulation.

## 5 Structural theory

**Definition 26** (Structural congruence)**.** $\equiv$ is the least congruence containing alpha-equivalence and satisfying associativity, commutativity and $\mathsf{0}$ as unit for $|$, together with

$$\ominus \ominus A \equiv A.$$

**Definition 27** (Name equivalence)**.** $\equiv_N$ is the least congruence satisfying

$$@_s(*_s x) \equiv_N i^k x, \qquad \frac{A \equiv B}{@_s A \equiv_N @_s B}$$

together with $@_s(\theta A) \equiv_N @_s(\theta' A)$, which keeps the payload coefficient out of a name's identity (Remark 45); here $k$ is the rotation of Definition 18. The first law is unconditional: under rotation every name is reconstructible from either drop, up to the phase the rotation introduced, so reflection holds at every name and not only on the diagonal.

*Remark* 28 (The completion equations, and where they live)*.* FORK, and the most consequential one in the note. A pair is intended as a formal difference, which suggests two further equations:

$$\langle Q, Q\rangle \equiv \mathsf{0}, \qquad \langle P, Q\rangle \equiv \langle P \mid R, Q \mid R\rangle.$$

These are what make the pair sort the group completion of processes rather than merely an ordered pair. But $\equiv_N$ contains $\equiv$ under quotation, and COMM fires on $\equiv_N$. Imposing them therefore makes them equations *on addresses*, which the matcher must decide.

The two readings differ sharply.

- **Arithmetic names.** The completion equations hold on names. Cancellation is then real, $\ominus$ is genuine negation, and destructive interference is the matcher discovering that a pair's components have become congruent. The cost is that unique factorisation is lost, the monomial basis with it, and the decidability of name matching becomes a serious question.

- **Coloured names.** The completion equations do not hold on names, which are then a free set with a $\mathbb{Z}/4$ action. Negation is a label carried along and read by the reduction rule, exactly as the sector is. The matcher is untouched; the arithmetic is deferred to a coefficient layer outside the calculus.

The reduction rule of Section 6 never asks whether two differently-signed names are equal, only whether they share an address, so the coloured reading suffices for the operational semantics as it stands. The arithmetic reading is what the semantic programme wants. The note takes neither; it records that the choice determines whether cancellation is a structural event or an external one.

# 6 Reduction

## 6.1 Matching

**Definition 29** (Matching)**.** A receipt at $x$ and a send at $x'$ are *matched* when $x' \equiv_N x$ or $x' \equiv_N \ominus x$.

So matching sees the sector, does not see the polarity, and does not see the payload coefficient, the last because Definition 27 keeps the coefficient out of a name's identity. Three consequences are worth stating before the rule uses them.

*Remark* 30 (What each blindness costs and buys)*. The sector is part of the address.* $@_l A$ and $@_r A$ are distinct names, and firing across them would fire on a relation strictly coarser than $\equiv_N$, which the calculus does nowhere else and which would break the injectivity of addressing on which unforgeability rests. It would also make the sender's sector unrecoverable from the reduct — discarded information rather than a phase.

*Polarity is not.* A name and its negation are one address. This is what allows a sign ever to be produced at a communication, and it is what Proposition 33 turns on.

*Neither is the coefficient.* Scaling a message must not move it to a channel that no receiver of the original is listening at, which is the requirement of Remark 16 applied one level up.

### 6.2 The rule

**Definition 31** (Relative polarity)**.** For matched $x$ and $x'$, the *relative polarity* $\delta(x, x')$ is the identity when $x' \equiv_N x$ and $\ominus$ when $x' \equiv_N \ominus x$. It is well defined because matched names agree in sector, so they differ by an element of the subgroup $\{1, \ominus\}$ of the $\mathbb{Z}/4$ torsor of Definition 21.

**Definition 32** (COMM)**.** Let $t := \mathsf{sec}(x)$. For matched $x, x'$,

$$\mathsf{for}(\theta'\langle\vec{y}_p, \vec{y}_n\rangle \leftarrow x)P \ \mid \ x'!\,\theta\langle\vec{Q}_p, \vec{Q}_n\rangle \ \rightarrow \ P\{\, N_i/y_{p,i}, \ \ \ominus N_i/y_{n,i}\,\}_i$$

where, writing $\delta := \delta(x, x')$,

$$N_i \ := \ @_t\big(F(\theta', \theta)\ \delta\langle Q_{p,i}, Q_{n,i}\rangle\big)$$

with $F$ the phase rule of Proposition 43. The rule is closed under parallel composition and under $\equiv$ in the usual way.

One clause, and four things to read off it.

**The sector is preserved.** The deposited quotation is in the sector the receipt listened on. No rule moves a name between sectors.

**The phase is declared, not computed.** The rotation applied to the deposited datum is $F(\theta', \theta)$, a function of what the receipt and the sender each declare, and it is the identity when both declare nothing. Section 7 derives $F$ and shows why no rule computing the rotation from the structure of the term could serve instead.

**Opposite polarity contributes the sign.** When the participants have opposite polarity, $\delta$ is $\ominus$ and the deposited pair is exchanged. The relative group element of the two participants is thus part of what is deposited, which is why one clause suffices where a rule distinguishing the polarity cases would need two.

**Each bound variable receives a formal difference.** $y_{p,i}$ receives $N_i$ and $y_{n,i}$ receives $\ominus N_i$: a rank-one name which is itself a difference, rather than a payload parked in one slot of an otherwise empty pair. Remark 34 shows this is forced.

### 6.3 Equivariance

**Proposition 33** (Equivariance)**.** *$P \rightarrow P'$ implies $\ominus P \rightarrow \ominus P'$.*

*Proof.* By Proposition 9, $\ominus$ carries the redex $\mathsf{for}(\langle\vec{y}_p, \vec{y}_n\rangle \leftarrow x)P \mid x'!A$ to $\mathsf{for}(\langle\vec{y}_n, \vec{y}_p\rangle \leftarrow \ominus x)\ominus P \mid (\ominus x')!(\ominus A)$, which is again a redex: matching is polarity-blind (Definition 29) and $\delta(\ominus x, \ominus x') = \delta(x, x')$. Reducing it binds $y_{n,i}$ to $@_t(\delta\langle Q_{n,i}, Q_{p,i}\rangle) = \ominus N_i$ and $y_{p,i}$ to $\ominus\ominus N_i = N_i$, which is the image under $\ominus$ of the original substitution, since $\ominus$ leaves the $N_i$ in the roles the exchanged binder assigns them. Hence both routes give $\ominus P\{N_i/y_{p,i}, \ominus N_i/y_{n,i}\}$, using $\ominus @_s B = @_s(\ominus B)$ and the commutation of $\ominus$ with substitution. $\square$

Equivariance is the design's selection principle: it is cheap to check and it decides between candidate rules that symmetry alone does not separate. Two things it settles.

*Remark* 34 (The deposit must be a whole pair). A rule depositing $@_t\langle Q, \mathsf{0}\rangle$ — the payload in the positive slot of an otherwise empty pair — is not equivariant. Nothing of that shape lies in the image of an equivariant rule, since $\ominus @_t\langle Q, \mathsf{0}\rangle = @_t\langle \mathsf{0}, Q\rangle$, and the failure is uniform at one application of $\ominus$ per communication. Depositing the whole pair, as Definition 32 does, is the unique repair.

**Proposition 35** (The deposit depends on the receipt's tag alone)**.** *Write the $\mathbb{Z}/4$ tags of Definition 21 additively and let $f(\alpha, \beta)$ be the tag deposited for a receipt tagged $\alpha$ served by a send tagged $\beta$. Equivariance requires $f(\alpha + 2, \beta + 2) = f(\alpha, \beta) + 2$, whose general solution is*

$$f(\alpha, \beta) \;=\; \alpha + h(\beta - \alpha)$$

*for some $h$. In particular the symmetric rule $f(\alpha, \beta) = \alpha + \beta$, under which the two tags multiply, is not available: it gives $f(\alpha + 2, \beta + 2) = f(\alpha, \beta) + 4 = f(\alpha, \beta)$.*

*Proof.* Negating both participants negates the deposit, which is the displayed condition; $f(\alpha, \beta) - \alpha$ is then invariant under adding 2 to both arguments, hence a function of $\beta - \alpha$ alone. □

*Remark* 36 (Which $h$). Proposition 35 leaves $h$ free, and matching restricts its argument to $\{0, 2\}$, so a rule of this shape has exactly two parameters. Section 7 shows that no constant choice of $h$ is satisfactory, which is why Definition 32 takes the rotation from the declared coefficients instead and leaves the sector alone.

# 7 Phase

Definition 32 defers the rotation to a function $F$ of two declared coefficients. This section says why the rotation cannot be computed from the term, derives $F$, and records what the naive form can and cannot do.

## 7.1 Why the rotation must be declared

**Proposition 37** (No constant rate)**.** *Let the deposited tag be $f(\alpha, r) = \alpha + h(r)$, where $r \in \{0, 2\}$ is the relative polarity of the two participants — the only freedom left once Section 36 is settled and matching is sector-sensitive. Then cancellation forces $h(2) - h(0) = 2$, and the two solutions with $h$ constant are exhaustive and both defective.*

*Proof.* For the constraint: consider $\mathsf{for}(Y \leftarrow x)P \mid x!A \mid (\ominus x)!(\ominus A)$, whose two candidates deposit $@_{\alpha+h(0)}A$ and $@_{\alpha+h(2)}(\ominus \ominus A) = @_{\alpha+h(2)}A$ — the same content. If $P$ does not use the datum the outcomes coincide, so their coefficients add, and a message composed with its own negation must sum to zero; hence the tags differ by two.

The two solutions are $h(0) = 1$ and $h(0) = 0$. With $h(0) = 1$ every communication rotates, so the sector of a name is the parity of the derivation length and two histories interfere only when their lengths agree; with $h(0) = 0$ no communication rotates, so a program begun in one sector remains there, and since matching is sector-sensitive the two sectors are non-interacting copies of the calculus and $i$ is inert. □

*Remark* 38 (The dilemma, and what it means). Proposition 37 is better read as an explanation than as an obstacle. *Rotation at communication is the only coupling between the sectors, and* *coupling them is exactly what makes phase accumulate with path length.* The reconvergence worry is therefore not an artefact of a convention: it is the price of $i$ doing anything at all, so long as the rate is a constant of the rule.

Both horns are removed by making the rotation a parameter of the site rather than of the rule. Ordinary same-polarity communication then carries no rotation by default, which restores easy reconvergence, while the sectors still couple wherever a program asks them to. Phase becomes *arranged* rather than *accrued*.

**Proposition 39** (The rotation cannot be derived from the payload)**.** *There is no function $\theta = \theta(A)$ of the payload alone that supports interference.*

*Proof.* Interference requires two candidates at one site reaching the same outcome with different phases. If the outcome is determined by the payload and the phase is a function of the payload, then candidates with coinciding outcomes have coinciding phases, so the relative phase at every recombination site is zero and nothing cancels. A derived phase is constant exactly where it must vary. □

*Remark* 40 (Rank is not a candidate either)*.* The one structural quantity that might have served is rank (Definition 2). It fails twice: it is additive under $\otimes$, so phase-equals-rank is a global function of the term, and a global function of the term is a gauge rather than a phase.

So the coefficient is a free parameter at the site, as it is in the ZX-calculus, where the phase decorates the spider. The remaining question is where to attach it.

## 7.2 The coefficient rides on the payload

**Definition 41** (Scaled payloads)**.** SETTLED. A payload is $\theta\langle\vec{P}, \vec{Q}\rangle$ with $\theta$ drawn from a phase group $\Theta$, and a receipt carries a coefficient $\theta' \in \Theta$ of its own. A payload so scaled is a *scaled formal difference*: a coefficient times a difference of two same-grade tuples, which is an element of a graded module over $\Theta$ written in the syntax.

*Remark* 42 (Why the payload and not the prefix)*.* A phase on the prefix is a property of the *act*; a phase on the payload is a property of the *thing sent*, and only the second survives being received. Under $x!_\theta A$ the coefficient is consumed at the site and the datum arrives bare, so a received name cannot carry its own amplitude and cannot be forwarded, stored or recombined with it. Under Definition 41 it can. A scalar has to travel with its vector.

**Proposition 43** (The phase rule)**.** *$F(\theta', \theta) = \theta' \cdot \theta$, the product in $\Theta$, and this is forced.*

*Proof.* Phases compose along a derivation, so $F$ must be a homomorphism in each argument; a trivial receipt against a trivial send must rotate nothing, so $F(1, 1) = 1$; and $\ominus$-equivariance is automatic, both coefficients being $\ominus$-invariant decorations (Remark 44). A bihomomorphism of groups preserving the unit is multiplication. □

*Remark* 44 (Coefficients are $\ominus$-invariant)*.* $\ominus(\theta A) := \theta(\ominus A)$. Were $\ominus$ to act on $\theta$ as well, the cancelling pair $x!\theta A$ and $(\ominus x)!\theta(\ominus A)$ would no longer differ by exactly two and the argument of Proposition 37 would fail at the site it exists to serve. So $-1$ is $\ominus$ and is not a value of $\theta$; see Remark 50.

*Remark* 45 (Coefficients are not part of an address). $@_s(\theta A) \equiv_N @_s(\theta' A)$: scaling a message must not move it to a channel that no receiver of the original is listening at, which is the same requirement that makes Definition 14 a shift rather than a scaling (Remark 16), applied one level up: multiplying a name by a name moves the address, so a coefficient must not be a factor in the content. The coefficient is carried as a decoration on the name, which is the status the sector was originally given and, by Remark 30, cannot have.

*Remark* 46 (Scaling and the multiplicatives). Definition 13 already multiplies the coefficients. Concatenation is different: tuples with different coefficients do not concatenate to a scaled tuple, so on payloads $\theta A +\!\!+\, \theta' B := (\theta\theta')(A +\!\!+\, B)$, with the coefficient pulled out in front rather than distributed slotwise. This is the syntactic shadow of a tensor of scaled vectors scaling by the product, and Definition 53 is to be read with it.

### 7.3 What the naive guard buys, and what it does not

The guard here is as simple as it can be: $\theta'$ is a constant of the receipt, not a formula evaluated against the candidate.

*Remark* 47 (Interference is arranged by senders). The obvious objection to a constant $\theta'$ is that it cannot give different candidates different phases, and phase-conditional-on-data is one of the two things interference requires. The objection is misplaced. At a contended channel the candidates differ by *which message wins*, and each message carries its own $\theta$; by Proposition 43 the deposited phases therefore vary across candidates by exactly the amount the senders declared. The receipt needs no formula language, no evaluation against the payload, and no purity argument, because the variation lives in the data rather than in the test.

**Proposition 48** (The rank-one ceiling)**.** *With $\theta'$ constant, the row of amplitudes at a receipt over the candidates $1, \ldots, m$ is $(\theta'\theta_1, \ldots, \theta'\theta_m)$, a scalar multiple of the vector of the senders' declarations. Every row of the clause matrix is proportional to that vector, so the matrix has rank one. No rank-one matrix is unitary except trivially, and none is basis-changing.*

*Remark* 49 (Where the ceiling sits). Proposition 48 is not an argument against the naive guard; it is a clean statement of its limit. The naive form suffices for phase composition, for the stochastic and probabilistic readings, and for cancellation — and it is insufficient for basis change, hence for universality. Recovering rank requires $\theta'$ to depend on the candidate, which is the non-naive guard: a formula evaluated against the payload, with the purity discipline that entails. The naive form is the default; the guard-computed $\theta'$ is the extension that buys rank.

*Remark* 50 (Where the sign lives). FORK, and it must be settled. There are now two possible carriers of a sign: the pair, through $\ominus$, and the coefficient, if $-1 \in \Theta$. Cancellation lands in neither unless they are identified,

$$@_s(\ominus A) \ \equiv_N \ -@_s A,$$

without which a message and the message with negated payload deposit structurally different names, their outcomes do not coincide, and nothing cancels: the sign becomes a label rather than an arithmetic fact.

But that equation makes the two carriers redundant. The recommendation of this note is that $\Theta$ be *sign-free* and the pair remain the sole carrier of $-1$, because cancellation must be structural: the matcher discovering $\langle Q, Q\rangle \equiv \mathtt{0}$ is what makes destructive interference an operational event rather than an entry in an external ledger. If instead $\Theta$ carries the sign, the matcher inherits arithmetic in the coefficient ring, which is Obligation 10 returning one level up and in a worse form.

# 8 Tensor and par

With sequences in hand there are two ways to combine two prefixes at two channels, and they are the two multiplicatives. Both keep the channels together and differ only in what happens to the binders: sharing a binder merges the two continuations' access to a single datum, which is context-merging; keeping the binders apart gives each continuation its own. The distinction is not available in a calculus with only $|$, and its availability here is what allows $\otimes$ and $⅋$ to be different operations rather than one degenerate one.

## 8.1 Par

**Definition 51** (Par of prefixes)**.**

$$\mathsf{for}(\vec{Y}^1 \leftarrow x)P_1 \ ⅋ \ \mathsf{for}(\vec{Y}^2 \leftarrow u)P_2 := \mathsf{for}(\vec{Y} \leftarrow x \cdot u)(P_1 ⅋ P_2)\{\vec{Y}/\vec{Y}^1, \vec{Y}/\vec{Y}^2\}$$
$$x!A \ ⅋ \ u!B := (x \cdot u)!(A ⅋ B)$$

where $\vec{Y}$ abbreviates a binder pair $\langle \vec{y}_p, \vec{y}_n \rangle$, the fresh $\vec{Y}$ has the common rank, and $A ⅋ B$ is slotwise. Substitution of one binder pair for another is polarity-respecting: positives for positives, negatives for negatives.

*Remark* 52 (Par is a comultiplication)*.* The identification of $\vec{y}^1$ and $\vec{y}^2$ with one $\vec{y}$ is a sharing, and sharing a datum between two continuations is copying it. Copying a basis element is the Frobenius comultiplication — in the graphical calculus, the spider that fans one wire out to two. So Definition 51 builds *correlated* composites, which is a legitimate and wanted operation, and it is not a tensor: see Remark 55.

## 8.2 Tensor

**Definition 53** (Tensor of prefixes)**.**

$$\mathsf{for}(\vec{Y}^1 \leftarrow x)P_1 \ \otimes \ \mathsf{for}(\vec{Y}^2 \leftarrow u)P_2 := \mathsf{for}(\vec{Y}^1 +\!\!+ \vec{Y}^2 \leftarrow x \cdot u)(P_1 \otimes P_2)$$
$$x!A \ \otimes \ u!B := (x \cdot u)!(A +\!\!+ B)$$

where $+\!\!+$ is concatenation, taken slotwise on pairs, so that binder pairs concatenate polarity by polarity. Both are extended by

$$(P_1 \mid P_2) \otimes Q := (P_1 \otimes Q) \mid (P_2 \otimes Q), \qquad \mathsf{0} \otimes Q := \mathsf{0}$$

and symmetrically in the second argument.

**Proposition 54** (Interchange)**.** *For $x, u$ of the appropriate ranks,*

$$\big(\mathsf{for}(\vec{Y}^1 \leftarrow x)P_1 \otimes \mathsf{for}(\vec{Y}^2 \leftarrow u)P_2\big) \ \mid \ \big(x!A \otimes u!B\big) \ \rightarrow \ P_1\{A/\vec{Y}^1\} \ \otimes \ P_2\{B/\vec{Y}^2\}$$

*modulo the phase of Definition 32,* provided *substitution distributes over $\otimes$.*

*Remark* 55 (Why the tensor needs separate binders). Under Definition 51 the same computation gives $P_1\{@(A \parr B)/\vec{Y}^1\} \parr P_2\{@(A \parr B)/\vec{Y}^2\}$: each continuation receives the *whole* combined payload rather than its own factor. Recovering the factors would require splitting a product of names, hence unique factorisation, which the completion equations of Remark 28 destroy. So interchange is exactly what separates the two operations, and it is the reason the tensor concatenates rather than shares.

*Remark* 56 (Rank is the tensor grade). Concatenation adds rank, so $\otimes$ grades the namespace, and the grading is free rather than symmetric because concatenation is not commutative. The symmetry of $\otimes$ is therefore an explicit permutation — a braiding, not strict commutativity — and symmetrisation or antisymmetrisation of a rank is a choice of twist rather than a property of the syntax. This does not reopen the Cayley–Dickson bound recorded above: that concerns the scalars, which remain commutative, so the calculus is still complex and not quaternionic.

### 8.3 Two consequences and two conjectures

**Proposition 57** (Slotwise par is not the addition)**.** *Let $A = \langle \vec{a}, \cdot \rangle$ and $B$ be of rank two and $C$ of rank one, and suppose addition of same-rank names were slotwise $|$. Then*

$$(A + B) \otimes C = \langle a_1 \mid b_1,\ a_2 \mid b_2,\ c\rangle, \qquad (A \otimes C) + (B \otimes C) = \langle a_1 \mid b_1,\ a_2 \mid b_2,\ c \mid c\rangle$$

*which differ: the untouched factor is duplicated. Hence $\otimes$ is not bilinear for slotwise $|$.*

**Corollary 58** (The sum in $\odot$ is external)**.** *Remark 75 is resolved against the internal reading. The linear structure must be formal sums over names as basis elements; slotwise parallel composition retains a role, but as the name* product *of Definition 13, not as the addition. Names accordingly carry two operations, $\cdot$ and $\otimes$, and $\otimes$ does not distribute over $\cdot$ — which is expected of a graded product and is not a defect.*

*Remark* 59 (Zero, and the missing unit). $\mathsf{0} \otimes Q = \mathsf{0}$ makes $\mathsf{0}$ the additive unit and the multiplicative annihilator at once, so $0 \cdot P = 0$ holds. That is worth noting because it is not true of the scalar action: $x \cdot \mathsf{0} \neq \mathsf{0}$ by Remark 16, and the collision between the two units — the additive unit of the content and the multiplicative unit of the namespace — was one of the standing defects of the unexpanded calculus. The other end is not resolved. $\otimes$ has no unit, since a unit would need channel content of rank zero and a body that is itself a unit, and no term serves for both prefix kinds. The structure is a non-unital graded rig, which is the same shape the join exhibits, and the fictitious identity step is its formal unitalisation.

**Conjecture 60** (Multiplicativity of evaluation)**.** $\Downarrow (P \otimes Q) = \Downarrow P \otimes \Downarrow Q$, equivalently

$$\langle P_1 \otimes P_2 \mid Q_1 \otimes Q_2\rangle = \langle P_1 \mid Q_1\rangle \cdot \langle P_2 \mid Q_2\rangle.$$

This is the criterion by which the operation earns its name, and it is what makes the failure of $\otimes$ to preserve single steps harmless. By distributivity a redex of $P$ appears once per component of $Q$ in $P \otimes Q$, so one step becomes a layer of steps and the image of $\otimes$ is not closed under $\to$; but the pairing looks only at terminal states, so non-closure in between costs nothing provided the endpoints agree. Two things follow if the conjecture holds: $(P \otimes Q)^\dagger = P^\dagger \otimes Q^\dagger$, and the states of Definition 82 are closed under $\otimes$, which is what makes it an operation on states rather than on terms.

**Conjecture 61** (De Morgan)**.** $(P \otimes Q)^\dagger \ = \ P^\dagger \parr Q^\dagger$.

If this holds the structure is $*$-autonomous rather than degenerately compact-closed, and the operator algebra ceases to be weaker than the state space. It is the criterion the two multiplicatives must jointly satisfy, and it is the reason for defining both rather than only the one the semantics immediately needs.

### 8.4 The heterogeneous cases

*Remark* 62 (Mixed prefix kinds are not defined, and should not be). SETTLED. $\otimes$ and $⅋$ are defined on like prefixes — receipt with receipt, send with send — and are undefined on a receipt against a send. That is not a gap. A receipt and a send at a common address are the two poles of an interaction, and their combination is the *pairing* of Definition 86, not a monoidal product: in the graphical calculus $\otimes$ is the product and the mixed case is the cup. The tensor-of-agents material in the earlier work restricts to what it calls homogeneous pairings for the same reason.

*Remark* 63 (Drops, and openness). $*_s x \otimes Q$ unfolds when $x$ is a concrete quotation and is stuck when $x$ is a bound variable, so $\otimes$ is total on closed terms and partial on open ones. This is the reflection obstacle in its usual place and is not removed by anything in this section.

*Remark* 64 (Different sectors, and a coherence argument). $@_l A \otimes @_r B$ multiplies channels of different sectors, hence multiplies their phases, hence uses precisely the table of Section 36. So the sector rule for the multiplicatives and the sector rule for COMM are one table, and the fork recorded there cannot be settled independently for the two. This is an internal coherence argument and is therefore stronger than the conservativity argument of Remark 67, which is about backward compatibility with a calculus this one does not claim to extend.

*Remark* 65 ($\otimes$ is not a congruence). After tensoring, the composite listens at $x \cdot u$ and no longer at $x$, so messages elsewhere in an ambient term addressed to $x$ can no longer reach it. $\otimes$ builds a new system rather than composing systems in place. This is the honest successor to the renaming device used in the earlier work to keep the two sides of a tensor from interacting: separation is now compositional address arithmetic rather than hygiene, which is a real improvement. It is still not *secure*, since $x \cdot u$ is computable by anyone; unforgeable separation continues to require a name-restriction operator.

## 9 Conservativity

**Proposition 66** (Embedding)**.** *FORK, stated as a target rather than a theorem. Let $\iota(-)$ send an ordinary* rho *term to the term of Definition 1 obtained by*

$$\iota(x!Q) := \iota(x)!\langle \iota(Q), \mathsf{0}\rangle, \qquad \iota(@P) := @_l\langle \iota(P), \mathsf{0}\rangle,$$
$$\iota(\mathsf{for}(y \leftarrow x)P) := \mathsf{for}(\langle y, y'\rangle \leftarrow \iota(x))\iota(P), \qquad \iota(*x) := *_l \iota(x)$$

*with $y'$ fresh, homomorphically elsewhere. Then $\iota(-)$ is injective and preserves $\equiv$.*

*Remark* 67 (The embedding is closed under reduction). Definition 32 preserves the sector and applies the identity rotation when neither participant declares a coefficient, so the image of $\iota(-)$ is closed under $\to$: an ordinary rho program embedded on the real axis stays there and reduces as it did.

This is a change from earlier versions, where the rule rotated the sector at every communication and the image was not closed — an ordinary program was not, in that calculus, a program that stayed classical. The present rule buys conservativity back, and Proposition 37 explains what it costs elsewhere: a rule that never rotates leaves the sectors uncoupled, which is what Obligation 6 is about.

# 10 Evaluation and residuation

The calculus above is a term language and a reduction relation. The construction it serves needs two further operations: an evaluation, which runs a composite to its terminal states and collects them, and a pole, against which the dagger is obtained by residuation. Both are recorded here. Following the design decision that the pole be taken as simply as possible for the time being, $\perp$ is the zero process and $P^{\perp}$ is residuation to it — the linear perp.

## 10.1 Inert states and evaluation

**Definition 68** (Inert)**.** $P$ is *inert* when $P \not\to$; that is, when no COMM applies to any of its components under $\equiv$. Write $\mathsf{Inert}$ for the inert processes.

Inertness is not triviality: $\mathsf{0}$ is inert, and so is $x!A$ with no matching receipt. The distinction matters throughout what follows, and is the reason $\perp$ below is not simply "is inert".

**Definition 69** (Evaluation product)**.**

$$P \odot Q \;:=\; \sum_{\substack{R \in \mathsf{Inert} \\ P|Q \,\to^{*}\, R}} R$$

where the sum ranges over terminal states with multiplicity, one summand per *causal history* ending at $R$.

**Proposition 70** (Commutativity)**.** *$P \odot Q \equiv Q \odot P$.*

*Proof.* $|$ is commutative and the index of the sum mentions $P$ and $Q$ only through $P \mid Q$. □

**Definition 71** (Evaluation)**.** $\Downarrow P := \mathsf{0} \odot P$.

*Remark* 72 (Histories, not paths)*.* SETTLED, and load-bearing rather than hygienic. Indexing the sum by reduction *sequences* rather than by causal histories counts the linearisations of a partial order, so $m$ independent redexes contribute a spurious factor of $m!$. The multiplicity in Definition 69 is therefore the number of histories, equivalently the number of maximal chains in the reduction order modulo permutation of independent steps. Concurrency squares are filled; what is summed is what survives the quotient.

*Remark* 73 (What $\odot$ is not)*.* Three properties fail, and each failure is informative rather than repairable.

*No unit.* $\mathsf{0} \odot P = \Downarrow P$, which is $P$ only when $P$ is already inert. So $\mathsf{0}$ is a unit on $\mathsf{Inert}$ and nowhere else, which is one reason the carrier of the eventual algebra must be cut down — see Section 10.2.

*Not idempotent.* $P \odot P \neq P$ in general, because two copies of a process interact. This is a feature: it is what makes multiplicity meaningful, and it is why the additive structure cannot be mixed choice, whose idempotence would force $2 = 1$.

*Not associative.* $(P \odot Q) \odot S$ forces $P \mid Q$ to reach a terminal state before $S$ arrives, so any global reduction of $P \mid Q \mid S$ that schedules a communication between $P$ and $S$ early, destroying a $P$–$Q$ redex, is a history of the unbracketed composite and of no bracketed one. The bracketed product undercounts, and the deficit is exactly the cross-redexes. Writing $M_R$ for the operator $\square \mid R$ and $T$ for one-step reduction, the deficit is the commutator $[T, M_R]$: the redexes between $R$ and the state that existed in neither alone. $M_R$ alone is diagonal in the occupation basis and commutes with every $M_S$; $T$ alone is the off-diagonal part; neither is interesting, and their failure to commute is where the content of the construction lives.

**Convention 74** (The $n$-ary form)**.** Because of the last item, the primitive is not the binary product but the family of symmetric $n$-ary evaluations

$$\odot(R_1, \ldots, R_n) \;:=\; \sum_{\substack{R \in \mathsf{Inert} \\ R_1 \mid \cdots \mid R_n \to^* R}} R$$

with one global evaluation and no bracketing. Symmetry under permutation is Proposition 70. Nothing needs to be associative because nothing is composed twice; factorisation of an $n$-ary evaluation into lower ones then becomes a property that holds exactly on causally independent components, rather than an axiom.

*Remark* 75 (What the sum is). FORK. Under the reading in which parallel composition is addition, the formal sum of Definition 69 is available in the calculus: it is the parallel composition of the terminal states. That reading is attractive, since it needs no new former and no tagging, and multiplicity is then counted by $\mid$ itself.

It has a hazard that must be checked before it is adopted. Inertness is not preserved by parallel composition: $x!A$ and $\mathsf{for}(\langle \vec{y}_p, \vec{y}_n \rangle \leftarrow x)P$ are each inert and their composition is not. So the parallel composition of the summands may reduce further, and the sum would not be a normal form. The alternative is an external formal sum over the multiset of terminal states, which is inert by fiat and costs a coefficient layer outside the calculus.

Two things are settled regardless. Tagging the summands with fresh names is *not* the way to defeat idempotence: a retained tag is a which-path record, distinguishing summands permanently and preventing exactly the coincidence of outcomes on which the addition of coefficients depends. And quotation carries $\mid$ to the name product, so quoting an internalised sum yields a *product* of quotes; whichever reading is taken, the sum must not be internalised beneath a quotation.

*Remark* 76 (Partiality). $\odot$ is partial and the sum may be infinite. Non-confluence is not itself an obstacle — the sum is over all terminal states, not over a chosen one — but divergence is: a composite with no terminal state contributes the empty sum, and a composite with infinitely many contributes a sum that requires a summability condition before it denotes anything. Both are recorded as Obligation 2.

## 10.2 The pole, and residuation

**Definition 77** (Residual)**.** For formulae $\varphi, \psi$ of the generated spatial logic, $\varphi \rhd \psi$ is the right adjoint to the separating conjunction:

$$P \models \varphi \rhd \psi \quad \text{iff} \quad \forall R.\, R \models \varphi \;\Rightarrow\; P \mid R \models \psi.$$

**Definition 78** (The pole)**.** $\perp$ is the behavioural formula

$$P \models \perp \quad \text{iff} \quad \Downarrow P \equiv \mathsf{0},$$

that is, every terminal state of $P$ is the zero process.

**Definition 79** (Linear perp)**.** $P^{\perp} := \{\, Q \;:\; P \odot Q \equiv \mathsf{0} \,\}$, and for a set $S$ of processes, $S^{\perp} := \bigcap_{P \in S} P^{\perp}$.

**Proposition 80** ($P^{\perp}$ is residuation to $\perp$)**.** *$Q \in P^{\perp}$ iff $Q \models \psi_P \triangleright \perp$, where $\psi_P$ is a formula whose extension is the equivalence class of $P$ in the logic.*

*Proof.* By Definition 77, $Q \models \psi_P \triangleright \perp$ iff $Q \mid R \models \perp$ for every $R$ in the extension of $\psi_P$; by Definition 78 that is $\Downarrow (Q \mid R) \equiv \mathsf{0}$, which is $R \odot Q \equiv \mathsf{0}$ by Definitions 69 and 71. □

**Corollary 81** (Symmetry)**.** *$Q \in P^{\perp}$ iff $P \in Q^{\perp}$.*

**Definition 82** (States)**.** $P$ is a *state* when $\{P\}^{\perp\perp} = \{P\}^{\perp\perp}$ is generated by $P$, that is, when $P$ is $\perp\perp$-closed. The states are the carrier of everything that follows.

*Remark* 83 (The closure is the null-space quotient)*.* $\{P\}^{\perp\perp}$ is the set of processes annihilated by every annihilator of $P$, so biorthogonal closure is closure under the testing preorder induced by the pole, with the tests being annihilators and success being reduction to the zero process. Three consequences.

It supplies the collapse that quotation cannot. Quotation is injective on $\mathsf{Proc}/\equiv$ and therefore identifies nothing; every identification in the eventual pairing comes from $\odot$ and from this closure, that is, from reduction.

It fixes the equivalence at which the construction lives, and the equivalence it fixes is *testing*, not bisimulation. Since $\equiv_N$ is intensional and testing is not, the coefficient object and the state space sit at different equivalences; that is a known tension and it is not resolved here.

It destroys freeness. $\mathsf{Proc}/\equiv$ under $\mid$ is free commutative on the non-parallel components, and closed sets are not freely generated. The occupation basis is a coordinate system for the unclosed calculus and does not survive into the closed one.

*Remark* 84 (Totality fails, and how it is repaired)*.* Not every process has an annihilator: a persistent or replicating process at a name no one else holds has none, so $\psi_P \triangleright \perp$ is unsatisfiable and the dagger is partial. Definition 82 is the repair adopted here — restrict to the closed elements, which is the standard move — and it is the reason the states are cut down rather than the pole loosened. The alternative, kept in reserve, is to relativise the pole to a residue, taking $R \vdash P \perp Q$ when every terminal state of $P \mid Q$ is $R$; that would make the pairing residue-valued in exactly the way the intended semantics wants, at the cost of a family of poles rather than one.

*Remark* 85 (Canonical witnesses are forced)*.* $P^{\perp}$ is a set, and the dagger must be a term, because it is going to be quoted. Quotation does not respect any behavioural equivalence, so a representative must be chosen and not merely existentially quantified. The choice adopted is by minimal structural complexity, the measure already present for names and processes, with ties broken by the summation of Definition 69 when the minimum is not unique.

### 10.3 The pairing

**Definition 86** (Dagger and pairing)**.** For a state $P$, $P^\dagger$ is the canonical witness of $\psi_P \rhd \bot$ in the sense of Remark 85, composed with the sector and polarity correction of Section 11. The pairing is

$$\langle P \mid Q\rangle \ :=\ @_l\langle P^\dagger \odot Q, \mathsf{0}\rangle$$

and, for a reified operator, using Convention 74,

$$\langle P \mid K \mid Q\rangle \ :=\ @_l\langle \odot\Big(P^\dagger,\, K^*_x,\, x!\langle Q, \mathsf{0}\rangle\Big)\,, \mathsf{0}\rangle$$

with $x$ private to the composite and $K^*_x := \mathsf{for}(\langle y, y'\rangle \leftarrow x)K[*_l y]$ the abstraction corresponding to $K$.

*Remark* 87 (Why the quotation is outermost, and why the operator is reified)*.* Two points, each answering an earlier error.

$\odot$ lands in $\mathsf{Proc}$ and the scalars are names, so the pairing must pass through a quotation to change sort; a definition that has $\odot$ produce a scalar directly collapses the two sorts at precisely the place where the collapse is dangerous. The quotation is applied once, after evaluation, and contributes no identifications of its own.

Reifying $K$ removes the bracketing question rather than answering it. Filling a hole and then evaluating, or evaluating and then filling, differ by the cross-redexes of Remark 73; passing the argument along a private wire means there is no hole-filling anywhere, and the bracket is manifestly the three-point evaluation it should have been. Privacy of $x$ is required, not cosmetic: it is what prevents $P^\dagger$ from intercepting the operator's wire, and renaming cannot supply it in a calculus where any party may construct any quotation.

*Remark* 88 (Annihilation is normalisation, not orthogonality)*.* An immediate and initially unwelcome consequence. $\langle P \mid P\rangle = @_l\langle \mathsf{0}, \mathsf{0}\rangle$, the unit of the name product, so annihilation gives 1. The pole is a *normalisation* condition: $P^\dagger$ is the process that reduces $P$ to the unit, and $\langle P \mid P\rangle = 1$ is a theorem rather than a stipulation. That is a gain, but it vacates the position orthogonality was expected to occupy, and the correspondence table of the earlier work, which reads "orthogonality is process annihilation", is backwards on this point.

Orthogonality must therefore come from cancellation among the summands of Definition 69, which requires additive inverses on those summands. This is precisely what the pair sort supplies and what an unexpanded namespace could not, its scalars being a free commutative monoid with no inverses: two histories reaching the same terminal state with opposite polarity cancel, and $\langle P \mid Q\rangle = 0$ is the vanishing of a sum of nonzero contributions rather than the absence of any. The expansion of the namespace and the definition of the pairing are answerable to each other at exactly this point.

## 11 The dagger, in one paragraph

The construction this calculus serves requires an antilinear involution on states. Section 10 supplies the pole and the residuation; what remains is the sector and polarity content, recorded here.

First, the dagger is not a syntactic polarity swap on term formers. It is obtained by residuation against a pole: $Q^\dagger$ is a witness of $\psi_Q \rhd \bot$, in the sense made precise by Proposition 80 and Remark 85. This makes the dagger a property first and a term second, hence stable under any presentation generating the same logic. The choice of $\perp$ as the zero process is the simplest available and is provisional; Remark 84 records the relativised alternative.

Second, the sector and polarity components of the dagger are now fixed. Flipping both sectors is the map $(a, b) \mapsto (b, a)$, which in the complex numbers is $z \mapsto i \cdot \bar{z}$ and not conjugation; taken as the dagger it makes the pairing skew-Hermitian and puts $\langle P \mid P \rangle$ in the imaginary sector, so positivity fails outright. The correction is to compose the sector flip with a pair exchange,

$$@_l \mapsto \ominus @_r, \qquad @_r \mapsto \ominus @_l,$$

which is $z \mapsto -i\bar{z}$, is involutive, and is assembled entirely from operations Definitions 7 and 22 already provide. Since Proposition 9 lifts $\ominus$ to processes, this composite acts on a whole process rather than name by name, which is what the pairing of Definition 86 requires and what an earlier presentation could not write down. Remark 11 remains in force: $\ominus$ supplies the dagger's polarity component and is not itself the dagger.

*Remark* 89 (The positivity obligation, made precise). Since every communication rotates the sector and the dagger rotates it once more, the sector in which $\langle P \mid P \rangle$ lands depends on the number of communications consumed in the annihilation of $P^\dagger$ against $P$, modulo four. Positivity therefore requires that all annihilation paths of $P^\dagger \mid P$ agree modulo four. This is a graded confluence condition, weaker than confluence, and it is not obviously true. It is cheap to test on the ladder of mutually annihilating names, and it should be tested before anything further is built on the dagger.

## 12 Obligations

**Obligation 1** (Cross-redexes and the unit of evaluation)**.** Remark 73 makes $\odot$ non-associative, with the deficit $[T, M_R]$. Convention 74 sidesteps the question for a single evaluation but does not answer it: composing operators, $\langle P \mid K_1 K_2 \mid Q \rangle$, reintroduces it in the form of whether the wires create cross-redexes. Determine the condition under which the $n$-ary evaluation factors, and check it against the reified operators of Definition 86.

**Obligation 2** (Summability)**.** Discharge Remark 76: give the condition under which the sum of Definition 69 denotes when the set of terminal states is infinite. Over a positive codomain monotone convergence suffices; with the pair sort supplying signs it does not, and an absolute-summability condition on the annihilation profile is the candidate.

**Obligation 3** (The sum, internal or external)**.** Settle Remark 75. If the sum is parallel composition, show that the composite of the terminal states is inert, or characterise when it is not. If external, say where the coefficient layer lives and how it interacts with Remark 28.

**Obligation 4** (Matching the logic to the pole)**.** Proposition 80 uses a formula $\psi_P$ whose extension is the equivalence class of $P$. If that logic separates more finely than $\perp\perp$-equivalence, then $P \mapsto P^\dagger$ does not factor through the states of Definition 82 and the dagger is ill defined on them. Either take the characteristic formula in a logic whose adequacy target is the pole's equivalence, or define the dagger on terms and prove separately that it descends.

**Obligation 5** (The four-case table)**.** DISCHARGED, and refuted: see Section 36. Equivariance under the lifted $\ominus$ requires $f(\alpha+2, \beta+2) = f(\alpha, \beta)+2$, which the symmetric reading fails. The conservativity argument previously recorded as the strongest one in its favour does not survive the collision, being an argument about backward compatibility with a calculus this one does not claim to extend.

**Obligation 6** (Do the sectors still earn their keep?)**.** THE LIVE QUESTION. With phase carried by the payload coefficient (Section 7) and the rotation supplied per candidate, Definition 32 preserves the sector, so no rule ever moves a name between sectors; and matching is sector-sensitive (Remark 30), so the two sectors do not communicate. By Proposition 37 that is the $h(0) = 0$ horn, under which the sectors are superselected and $i$ is inert as a feature of the namespace.

What remains load-bearing is the *pair*, which carries the sign structurally and is what makes cancellation an event the matcher can discover. What may now be idle is the doubling of the quotation: $i$ is a value of $\Theta$, phase composition is multiplication in $\Theta$ (Proposition 43), and the $\mathbb{Z}/4$ coupling of Definition 22 — the one stipulation in the whole design, and the least comfortable thing in this note — would be unnecessary.

The test is cheap and should be run before any further development: write down what breaks if $@_l$ and $@_r$ are identified while the pair is kept. Sections 4 and the sector clauses of Definitions 18, 27 and 32 are what would collapse.

**Obligation 7** (Rank, and the non-naive guard)**.** Proposition 48 caps the naive guard at rank one, hence below basis change. Specify the guard-computed $\theta'$ — a formula evaluated against the candidate, with the purity discipline that entails — and determine what rank it reaches. This is the gap between a calculus that composes phases and one that computes with them.

**Obligation 8** (The phase group)**.** Fix $\Theta$. Remark 50 recommends that it be sign-free, with the pair carrying $-1$; that recommendation has to be checked against the requirement that $\Theta$ contain enough phases to be useful, since a sign-free group containing $i$ has $i^2 = -1$ in it whether or not the syntax admits the value. Settle also whether $\Theta$ is a group or a rig, and whether the formal sums of Corollary 58 take coefficients in $\Theta$ or in something larger.

**Obligation 9** (Arithmetic or coloured names)**.** Settle Remark 28: whether the completion equations hold on names, and hence whether cancellation is a structural event decided by the matcher or an external one deferred to a coefficient layer.

**Obligation 10** (Decidability of name matching)**.** COMM fires on $\equiv_N$. Under the arithmetic reading of Remark 28, the matcher must normalise formal differences nested through quotation. Termination is not obvious, and this is the practical gate on the entire design.

**Obligation 11** (Freshness)**.** The property that a quotation of $P$ is not free in $P$ is what obviates a freshness operator and underwrites freshness by quotation. Its proof proceeds by structural complexity. Both quotation operators must increase that measure, which is routine; but the quote-drop law of Definition 27 relates a name to a term built from its own drops, and the completion equations, if adopted, shorten terms non-structurally. The mutual recursion of $\equiv$ and $\equiv_N$ was already the delicate part of the metatheory. That argument must be redone rather than adapted.

**Obligation 12** (Observability of the sector)**.** DISCHARGED by Definition 18, and recorded because it was open in earlier versions. When the cross cases of the drops projected to $\mathsf{0}$, a context could apply both drops and read the sector, making it a superselection rule; the remedy contemplated was to restrict drops to guard position. Under rotation both drops succeed and differ by a phase, so the sector is not readable by comparing them and no restriction on where drops may occur is needed. See Remark 19.

**Obligation 13** (Reprove the collapse proposition)**.** Under Definition 18 the two drops are interdefinable, so Proposition 17 no longer concerns independent term formers. Remark 20 argues that the substance survives; the proof should be redone rather than assumed.

**Obligation 14** (Substitution and the tensor)**.** Proposition 54 is conditional on substitution distributing over $\otimes$. Discharge or refute it; the interchange law, and with it the distinction between the two multiplicatives, turns on this.

**Obligation 15** (The two conjectures)**.** Settle Conjectures 60 and 61. The first decides whether $\otimes$ deserves its name and whether the states are closed under it; the second decides whether the structure is $*$-autonomous or degenerately compact-closed.

**Obligation 16** (Rank and arity)**.** Definition 2 gives a name a rank and a prefix an arity, and Definitions 13 and 53 keep them apart: channels multiply, so their contents combine slotwise and their rank is unchanged, while payloads concatenate and arity adds. Whether the two numbers should instead be identified — which would require channels to concatenate as well, making $\otimes$ uniformly concatenation — is open, and it is the cleanest remaining question about the graded structure.

**Obligation 17** (Scalar extraction and occurrence counts)**.** For a race to interfere, its candidates must reach the same outcome with different coefficients. Consuming from $@_l A$ rather than $@_r A$ yields $P\{@_r Q/y\}$ rather than $P\{@_l Q/y\}$, and these are the same outcome up to phase only if the sector can be extracted from the name to the term, that is, only if there is a homomorphism from processes to $\mathbb{Z}/4$ under which substitution multiplies. The natural candidate is sector degree, the product of sectors over name occurrences — but then the phase is raised to the power of the number of occurrences of $y$, so extraction holds only when $y$ occurs exactly once. Occurrence count is the exponent on the phase. Either linear use of names bound from graded channels is required, or multiplicative phase in occurrences is accepted and scalar extraction abandoned. This is a linearity requirement of a different kind from the ones previously considered and declined, and the earlier reasons for declining do not apply to it.

**Obligation 18** (Positivity)**.** Discharge or refute Remark 89: that all annihilation paths of $P^\dagger \mid P$ agree in length modulo four.

**Obligation 19** (Two gradings on names)**.** This calculus grades names by sector. A separate line of work grades them by binding structure, unforgeable names being those introduced by a name-restriction operator and classical ones being quotations. The two gradings are independent, so there are four kinds of name, and their interaction is unstated. The conjecture worth testing is that sectors should live only in the unforgeable namespace, which would also confine Obligation 12.

**Obligation 20** (Metatheory)**.** Nothing in this note is proved beyond Propositions 8, 17 and 23. Subject reduction for the sector grading, whether bisimulation is a congruence in the presence of two quotations, and whether the sector is visible to bisimulation at all, are all open. The last is pressing: an inert distinction that $\equiv$ can see and $\sim$ cannot is exactly the configuration in which the spatial layer strictly refines behaviour, and the sign structure would then be behaviourally invisible.

## 13 What this note does not do

It does not give the interpretation map, and it does not establish that the resulting scalars form a rig — only that of the four obstructions to their doing so, one is discharged and the other three have each been given a place to be repaired. The evaluation and the pole of Section 10 are stated, not developed: nothing is proved about $\odot$ beyond commutativity, the pole is the simplest one available and is provisional, and the summability question of Obligation 2 is open, so the pairing of Definition 86 is a definition and not yet a well-defined operation. It does not treat the join, whose generalisation to graded messages requires a sum over matchings with coefficients in a group ring, and whose normalisation may or may not remain the right unit. And it takes no position on whether the resulting calculus is worth the cost of Obligation 10, which is the question a reader should keep in view throughout.

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